Statistics for International Commerce

Week 4: Discrete Probability Distributions

Iegor Vyshnevskyi, Ph.D.

Sogang University, Geppert School of International Studies

March 24/26, 2026

Hello everyone!

It's a **great day** to *keep learning* statistics for international commerce. :-)

Agenda

- Probability distributions
- Random variables
- Binomial distribution
- Hypergeometric distribution
- Poisson distribution
- In-class exercises

Learning Objectives

After this lecture students should be able to:

- Understand probability distributions
- Distinguish discrete vs continuous variables
- Compute expected value of a distribution
- Apply binomial, hypergeometric, and Poisson distributions
- Implement probability models in Python

What is a Probability Distribution?

A probability distribution describes:

- all possible outcomes of a random experiment
- the probability of each outcome

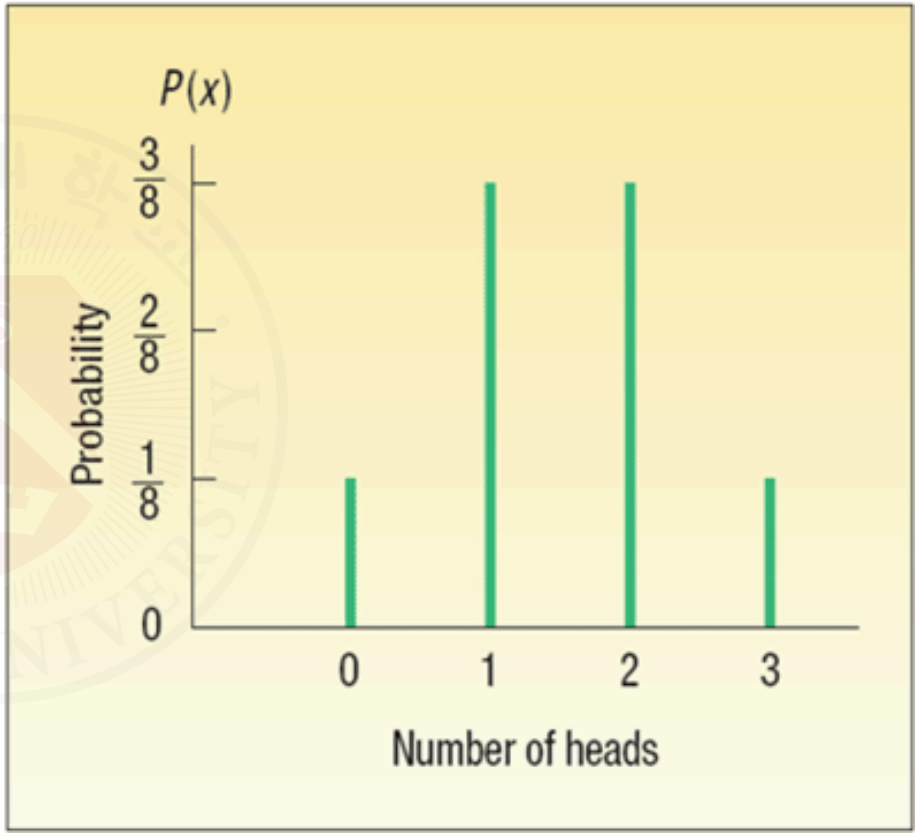
Experiment: Toss a coin three times. Observe the number of heads. The possible results are: zero heads, one head, two heads, and three heads.

What is the probability distribution for the number of heads?

Possible Result	Coin Toss			Number of Heads
	First	Second	Third	
1	T	T	T	0
2	T	T	H	1
3	T	H	T	1
4	T	H	H	2
5	H	T	T	1
6	H	T	H	2
7	H	H	T	2
8	H	H	H	3

Source: Douglas Lind, William Marchal, Samuel Wathen, Statistical Techniques in Business and Economics, 16th ed. (LMW)

Number of Heads, x	Probability of Outcome, $P(x)$
0	$\frac{1}{8} = .125$
1	$\frac{3}{8} = .375$
2	$\frac{3}{8} = .375$
3	$\frac{1}{8} = .125$
Total	$\frac{8}{8} = 1.000$



Source: Douglas Lind, William Marchal, Samuel Wathen, Statistical Techniques in Business and Economics, 16th ed. (LMW)

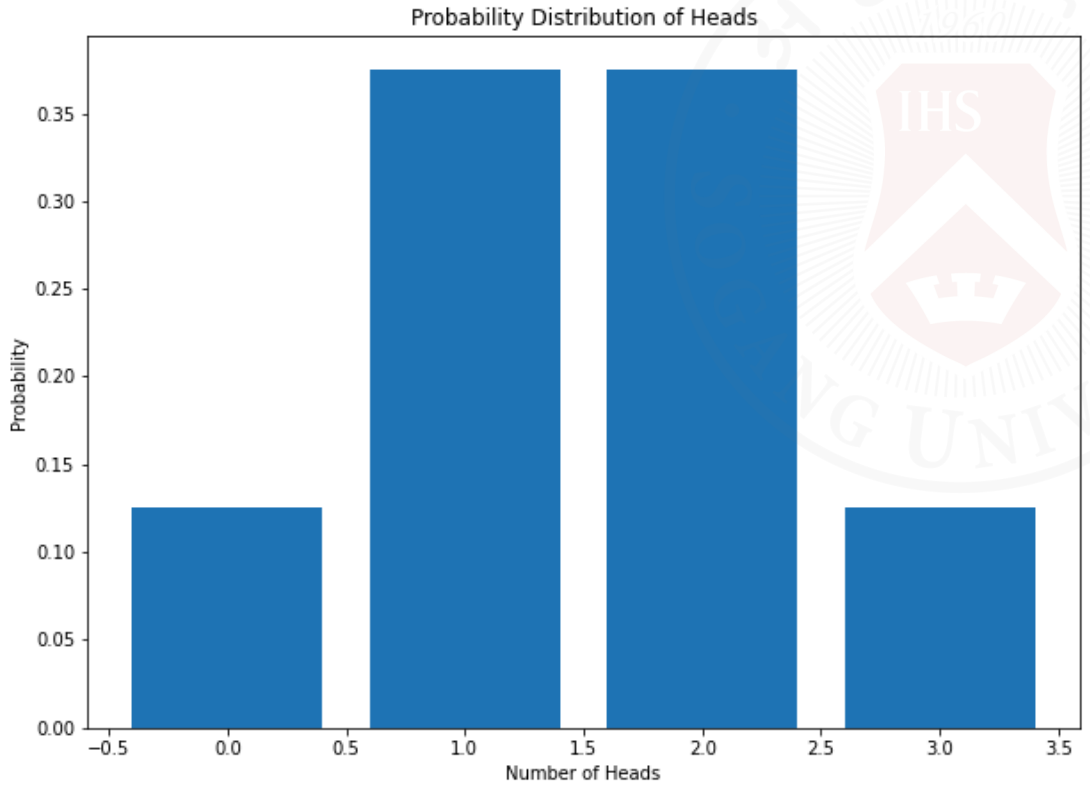
Python Example: Coin Toss Distribution

```python
import pandas as pd

data = {
    "Heads": [0,1,2,3],
    "Probability": [1/8,3/8,3/8,1/8]
}

df = pd.DataFrame(data)

df
```

```
##    Heads  Probability
## 0      0        0.125
## 1      1        0.375
## 2      2        0.375
## 3      3        0.125
```

Visualizing distribution

```
import matplotlib.pyplot as plt

plt.bar(df["Heads"], df["Probability"])
plt.title("Probability Distribution of Heads")
plt.xlabel("Number of Heads")
plt.ylabel("Probability")
plt.show()
```



Random Variables

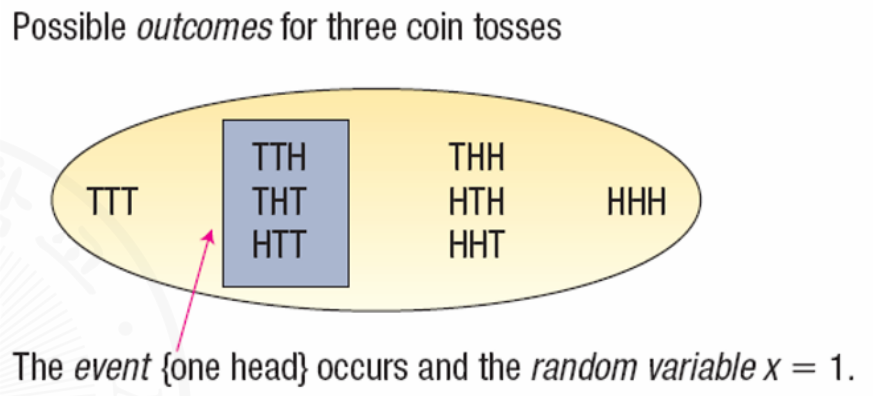
A random variable is a numerical value resulting from a random experiment.

Example

Number of delayed flights per day.

Possible values

0, 1, 2, 3, ...



Source: Douglas Lind, William Marchal, Samuel Wathen, Statistical Techniques in Business and Economics, 16th ed. (LMW)

Types of Random Variables

Discrete random variable

- Countable outcomes
- Example: number of export shipments delayed, number of cars sold, number of customers arriving

Continuous random variable

- Measured outcomes
- Example: shipping time, exchange rate changes, customer satisfaction score

> Intuition: Discrete = countable, Continuous = measurable.

Features of a Discrete Distribution

The main features of a discrete probability distribution are:

- The sum of the probabilities of the various outcomes is 1.00.
- The probability of a particular outcome is between 0 and 1.00.
- The outcomes are mutually exclusive.

The Mean of a Probability Distribution

Expected Value (Mean)

- The mean is a typical value used to represent the central location of a probability distribution.
- The mean of a probability distribution is also referred to as its expected value.

Formula

MEAN OF A PROBABILITY DISTRIBUTION $\mu = \Sigma[xP(x)]$ **[6–1]**

Source: Douglas Lind, William Marchal, Samuel Wathen, Statistical Techniques in Business and Economics, 16th ed. (LMW)

Interpretation: The expected value is the average outcome of a large number of repeated experiments.

Example: Expected number of cars sold on Saturday

Example: Expected Sales Distribution

Cars sold on Saturday

$$\begin{aligned}\mu &= \Sigma[xP(x)] \\ &= 0(.10) + 1(.20) + 2(.30) + 3(.30) + 4(.10) \\ &= 2.1\end{aligned}$$

Number of Cars Sold, x	Probability, $P(x)$	$x \cdot P(x)$
0	.10	0.00
1	.20	0.20
2	.30	0.60
3	.30	0.90
4	.10	0.40
Total	1.00	$\mu = 2.10$

Source: Douglas Lind, William Marchal, Samuel Wathen, Statistical Techniques in Business and Economics, 16th ed. (LMW)

Expected value

$$\mu = \Sigma x P(x) = 2.1$$

This means the salesperson sells about **2 cars on average**.

Python Example: Expected Value

```python
import pandas as pd

data = {
    "cars_sold":[0,1,2,3,4],
    "prob":[0.10,0.20,0.30,0.30,0.10]
}

df = pd.DataFrame(data)

expected_value = (df["cars_sold"] * df["prob"]).sum()

print(f"Expected value: {expected_value}")
```

```
## Expected value: 2.1
```

Variance and Standard Deviation

Variance measures spread of outcomes.

Formula

$$\sigma^2 = \Sigma(x - \mu)^2 P(x)$$

Standard deviation

$$\sigma = \sqrt{variance}$$

Variance and Standard Deviation

Number of Cars Sold, x	Probability, $P(x)$	$(x - \mu)$	$(x - \mu)^2$	$(x - \mu)^2 P(x)$
0	.10	0 − 2.1	4.41	0.441
1	.20	1 − 2.1	1.21	0.242
2	.30	2 − 2.1	0.01	0.003
3	.30	3 − 2.1	0.81	0.243
4	.10	4 − 2.1	3.61	0.361
				$\sigma^2 = 1.290$

Source: Douglas Lind, William Marchal, Samuel Wathen, Statistical Techniques in Business and Economics, 16th ed. (LMW)

$$\text{Variance} = \Sigma(X - \mu)^2 * P(X) = 1.296$$

$$\text{Standard Deviation} = \sqrt{1.296} = 1.138$$

Python Example: Variance of Distribution

```python
import numpy as np
mu = expected_value
variance = ((df["cars_sold"] - mu)**2 * df["prob"]).sum()
std_dev = np.sqrt(variance)
print(f"Variance: {variance}")
```

```
## Variance: 1.29
```

```python
print(f"Standard Deviation: {std_dev}")
```

```
## Standard Deviation: 1.1357816691600546
```

Binomial Distribution

> A binomial distribution describes the number of successes in repeated trials.

Characteristics of Binomial Distribution

- Two outcomes (success or failure)
- Fixed number of trials
- Independent trials
- Same probability each trial

Examples:

- Number of delayed shipments in a day
- Number of customers who make a purchase

Binomial Formula

BINOMIAL PROBABILITY FORMULA $$P(x) = {}_nC_x\, \pi^x (1 - \pi)^{n-x} \qquad [6\text{–}3]$$

where:

C denotes a combination.
n is the number of trials.
x is the random variable defined as the number of successes.
π is the probability of a success on each trial.

Source: Douglas Lind, William Marchal, Samuel Wathen, Statistical Techniques in Business and Economics, 16th ed. (LMW)

Example: Late Flights

Suppose:

- 5 flights per day
- Probability of delay = 0.20

Question: What is probability that **no flights are late?**

$$
\begin{aligned}
P(x=0) &= {}_nC_x \pi^x (1-\pi)^{n-x} \\
&= {}_5C_0 (.20)^0 (1\text{-}.20)^{5\text{-}0} \\
&= (1)(1)(.3277) \\
&= 0.3277
\end{aligned}
$$

Source: Douglas Lind, William Marchal, Samuel Wathen, Statistical Techniques in Business and Economics, 16th ed. (LMW)

Python Example: Binomial Distribution

```python
from scipy.stats import binom

n = 5
p = 0.20

prob_no_delay = binom.pmf(0, n, p)

print(f"Probability of no delays: {prob_no_delay:.4f}")
```

```
## Probability of no delays: 0.3277
```

Mean and Variance of Binomial Distribution

Mean

$$\mu = np$$

Variance

$$\sigma^2 = np(1-p)$$

Example

- n = 5
- p = 0.2
- Mean = 1
- Variance = 0.8

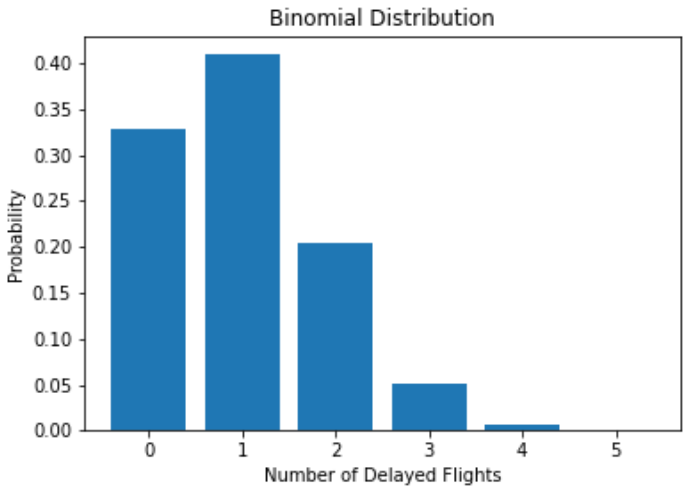
$$\mu = n\pi$$
$$= (5)(0.20) = 1.0$$

$$\sigma^2 = n\pi(1-\pi)$$
$$= (5)(0.20)(1-0.20)$$
$$= (5)(0.20)(0.80)$$
$$= 0.80$$

Source: Douglas Lind, William Marchal, Samuel Wathen, Statistical Techniques in Business and Economics, 16th ed. (LMW)

Visualizing Binomial Distribution

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import binom
x = np.arange(0,6)
y = binom.pmf(x,5,0.2)
plt.bar(x,y)
plt.title("Binomial Distribution")
plt.xlabel("Number of Delayed Flights")
plt.ylabel("Probability")
plt.show()
```



Binomial Dist. – Mean and Variance: Another Solution

Number of Late Flights, x	$P(x)$	$xP(x)$	$x - \mu$	$(x - \mu)^2$	$(x - \mu)^2 P(x)$
0	0.3277	0.0000	−1	1	0.3277
1	0.4096	0.4096	0	0	0
2	0.2048	0.4096	1	1	0.2048
3	0.0512	0.1536	2	4	0.2048
4	0.0064	0.0256	3	9	0.0576
5	0.0003	0.0015	4	16	0.0048
		$\mu = 1.0000$			$\sigma^2 = 0.7997$

Source: Douglas Lind, William Marchal, Samuel Wathen, Statistical Techniques in Business and Economics, 16th ed. (LMW)

Binomial Distribution - Table

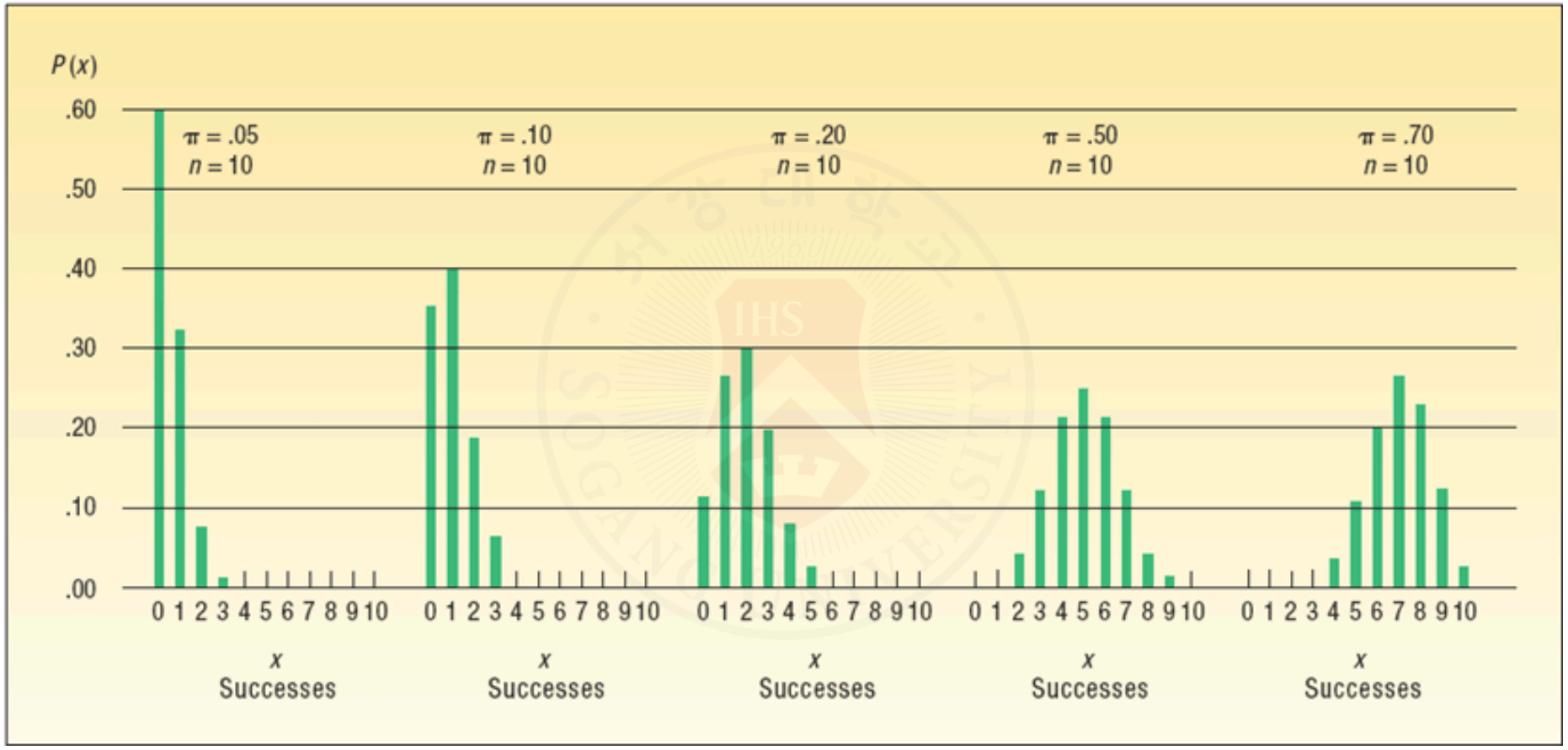
Five percent of the worm gears produced by an automatic, high-speed Carter-Bell milling machine are defective. What is the probability that out of six gears selected at random none will be defective? Exactly one? Exactly two? Exactly three? Exactly four? Exactly five? Exactly six out of six?

TABLE 6–2 Binomial Probabilities for $n = 6$ and Selecte Values of π

	$n = 6$ Probability										
$x \backslash \pi$	.05	.1	.2	.3	.4	.5	.6	.7	.8	.9	.95
0	.735	.531	.262	.118	.047	.016	.004	.001	.000	.000	.000
1	.232	.354	.393	.303	.187	.094	.037	.010	.002	.000	.000
2	.031	.098	.246	.324	.311	.234	.138	.060	.015	.001	.000
3	.002	.015	.082	.185	.276	.313	.276	.185	.082	.015	.002
4	.000	.001	.015	.060	.138	.234	.311	.324	.246	.098	.031
5	.000	.000	.002	.010	.037	.094	.187	.303	.393	.354	.232
6	.000	.000	.000	.001	.004	.016	.047	.118	.262	.531	.735

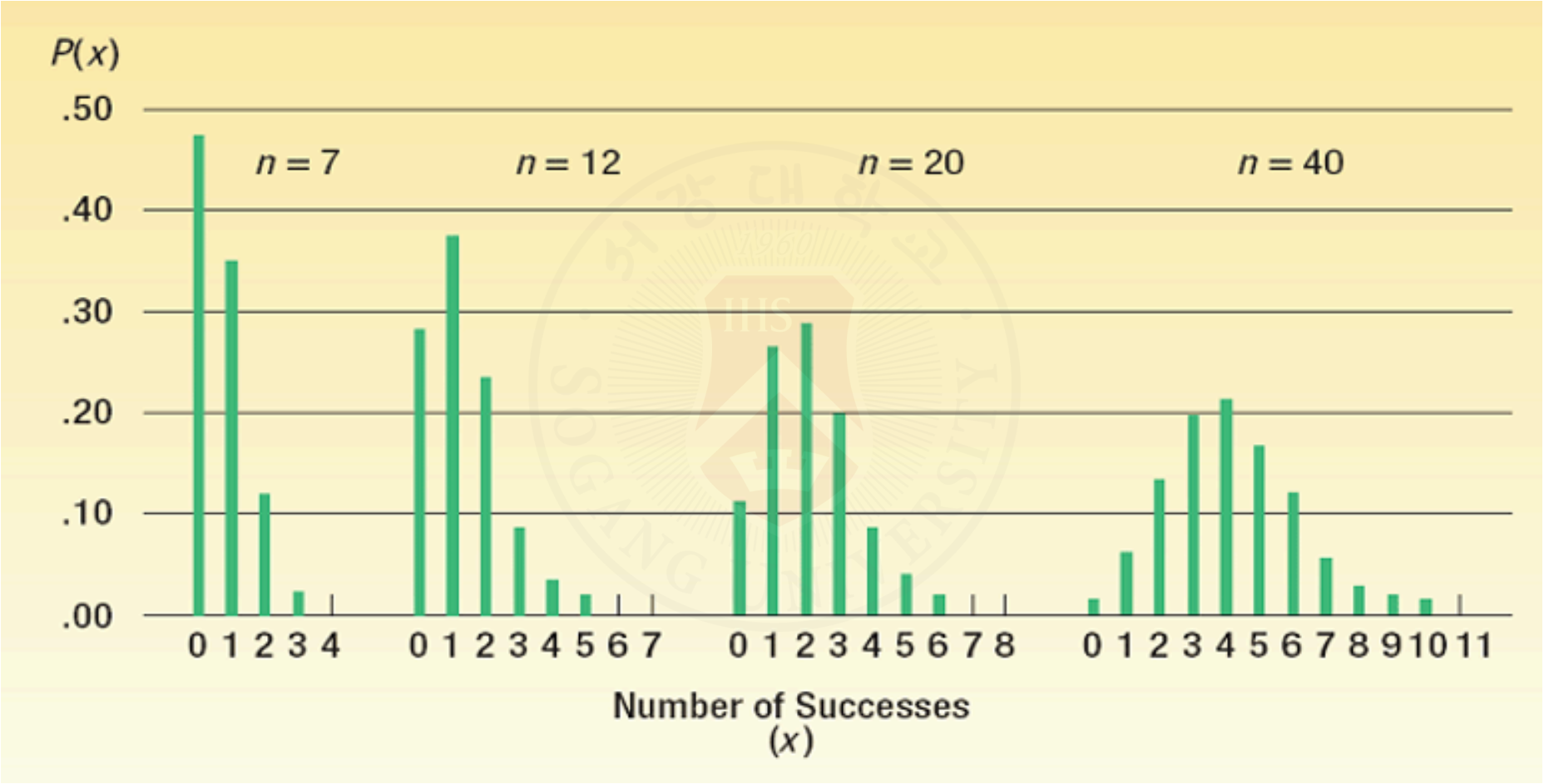
Source: Douglas Lind, William Marchal, Samuel Wathen, Statistical Techniques in Business and Economics, 16th ed. (LMW)

Binomial - Shapes for Varying \pi (n constant)



Source: Douglas Lind, William Marchal, Samuel Wathen, Statistical Techniques in Business and Economics, 16th ed. (LMW)

Binomial - Shapes for Varying n (constant)



Source: Douglas Lind, William Marchal, Samuel Wathen, Statistical Techniques in Business and Economics, 16th ed. (LMW)

Binomial Distribution - Python Example

```python
from scipy.stats import binom
# Parameters
n = 6  # number of trials
p = 0.05  # probability of success
# Probability of exactly 0 successes
prob_0 = binom.pmf(0, n, p)
# Probability of exactly 1 success
prob_1 = binom.pmf(1, n, p)
# Probability of exactly 2 successes
prob_2 = binom.pmf(2, n, p)
# Probability of exactly 3 successes
prob_3 = binom.pmf(3, n, p)
# Probability of exactly 4 successes
prob_4 = binom.pmf(4, n, p)
# Probability of exactly 5 successes
prob_5 = binom.pmf(5, n, p)
# Probability of exactly 6 successes
prob_6 = binom.pmf(6, n, p)

# Print results
print(f"Probability of exactly 0 successes: {prob_0:.4f}")
```

```
## Probability of exactly 0 successes: 0.7351
```

```python
print(f"Probability of exactly 1 success: {prob_1:.4f}")
```

```
## Probability of exactly 1 success: 0.2321
```

```python
print(f"Probability of exactly 2 successes: {prob_2:.4f}")
```

```
## Probability of exactly 2 successes: 0.0305
```

Cumulative Binomial Probability Distributions

A study in June 2003 by the Illinois Department of Transportation concluded that 76.2 percent of front seat occupants used seat belts. A sample of 12 vehicles is selected. What is the probability the front seat occupants in at least 7 of the 12 vehicles are wearing seat belts?

$$
\begin{aligned}
P(X \geq 7 | n = 12 \text{ and } \pi = .762) \\
&= P(X=7) + P(X=8) + P(X=9) + P(X=10) + P(X=11) + P(X=12) \\
&= .0902 + .1805 + .2569 + .2467 + .1436 + .0383 \\
&= .9563
\end{aligned}
$$

$$
\begin{aligned}
P(x = 7 | n = 12 \text{ and } \pi = .762) \\
&= {}_{12}C_7(.762)^7(1 - .762)^{12-7} = 792(.149171)(.000764) = .0902
\end{aligned}
$$

Source: Douglas Lind, William Marchal, Samuel Wathen, Statistical Techniques in Business and Economics, 16th ed. (LMW)

Other distributions: Hypergeometric Distribution

> Used when sampling **without replacement**.

Example

A factory has defective and non-defective items.

Selecting items without replacement changes probabilities.

Example: Union Employees

- Total workers = 50
- Union members = 40
- Non-union = 10

Select 5 employees.

Probability that **4 belong to union**.

Python Example: Hypergeometric

```python
from scipy.stats import hypergeom

N = 50   # population
K = 40   # success in population
n = 5    # sample size
x = 4    # successes in sample

print(f"Probability of {x} successes: {hypergeom.pmf(x, N, K, n):.4f}")
```

```
## Probability of 4 successes: 0.4313
```

Other distributions: Poisson Distribution

> Used to model rare events.

Examples

- lost baggage
- system failures
- arrivals of customers

Parameter

λ = average number of events.

Poisson Example

Average lost baggage per flight

$\lambda = 0.3$

Probability of **zero lost bags**

Python Example: Poisson Distribution

```python
from scipy.stats import poisson
lam = 0.3
prob_zero = poisson.pmf(0, lam)
print(f"Probability of zero lost bags: {prob_zero:.4f}")
```

```
## Probability of zero lost bags: 0.7408
```

Why Probability Distributions Matter in Economics

Applications

- trade risk analysis (e.g., probability of demand surge)
- logistics reliability (e.g., probability of shipment delay)
- financial default risk (e.g., probability of loan default)
- quality control in production (e.g., probability of defective items)
- forecasting demand (e.g., probability of high demand)

In-Class Exercise

A company ships goods internationally.

Probability shipment delayed = 0.15

If 8 shipments occur today

Questions

1. Probability exactly 2 delays
2. Probability at least one delay

Please compute using Python.

Python Exercise

```python
from scipy.stats import binom

n = 8
p = 0.15

print(f"Probability of exactly 2 delays: {binom.pmf(2, n, p):.4f}")
```

```
## Probability of exactly 2 delays: 0.2376
```

```python
print(f"Probability of at least one delay: {1 - binom.pmf(0, n, p):.4f}")
```

```
## Probability of at least one delay: 0.7275
```

Practice Session

Open the **Week 4 practice notebook** in LMS.

Complete probability exercises using Python.

Key Takeaways

- Probability distributions model uncertainty in commerce.
- Binomial, hypergeometric, and Poisson are key discrete distributions.
- Python allows practical application of these models.

Next Week

(Mar 31 | April 2) Continuous probability distributions (LMW Chapter 7) & **April 2 is a public holiday**. Video lecture will be uploaded.

Please submit by next week (Tuesday) your:

- Class activity (Colab link).

Any questions?

Thank you for your attention and active participation!